

we evolve a probability distribution ρ_k

gradient (for explicit) distanc (for implicit)

From $\mathbb{R}^d$ to $\mathcal{P}(\mathbb{R}^d)$

- What if instead of a single point x_t , **we evolve a probability distribution ρ_k** ?
- Let $F : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$ be a functional over distributions, ρ_0 a starting point
- Can we define a gradient flow of F ?
- I.e., curve in $\mathcal{P}(\mathbb{R}^d)$ starting at ρ_0 and evolving in the direction of maximal descent of F ?
- We need: notion of **gradient (for explicit)** or notion of **distance (for implicit)**
- Much easier to start from distance!

From $\mathbb{R}^d$ to $\mathcal{P}(\mathbb{R}^d)$